# Bridging Runtime Gaps in LLVM: Vendor-Agnostic Dispatch for ML Kernels

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Poster Session

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## The Gap

**MLIR compiles one `gpu.module` to 3+ GPU vendors — but picks the first compatible binary at runtime.**

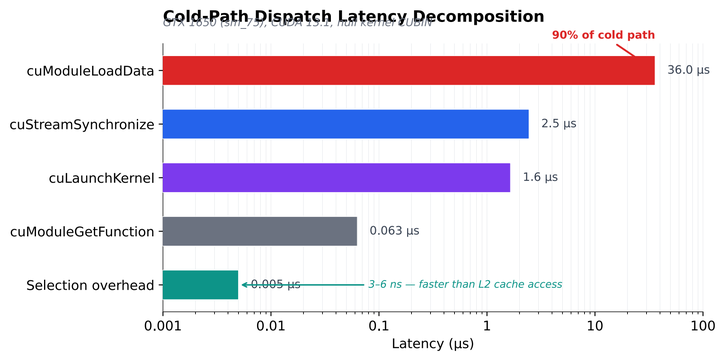
**OffloadBinary** carries N device images. The runtime loads the first image that doesn't fail. No metadata vocabulary. No measurement. No "best-compatible" mechanism.

## Upstream Evidence

| PR / Issue | What it shows |
|---|---|
| #148286 | XeVM — new vendor images arriving fast |
| #186088 | liboffload uses first-wins selection |
| #185663 | `isMetadataCompatible` — no policy |
| #75356 | Chapel users need dispatch |
| #88170 | RFC: policy slot explicitly empty |

5 independent signals pointing at the same gap.

## Phase 1: Dispatch Measurement




| 36 µs | 3--6 ns | < 0.02% |
|---|---|---|
| cuModuleLoadData (90% of cold path) | Selection overhead | Dispatch cost vs. kernel load |

**Selection is free relative to driver costs. So: what information should drive it?**

## Metadata Vocabulary

| Key | Purpose |
|---|---|
| `min_sm` | Min CUDA compute capability |
| `min_gfx` | Min AMD GFX version |
| `requires_features` | Tensor cores, matrix units, etc. |
| `variant_priority` | Explicit ordering for tie-breaking |
| `variant_tag` | Human-readable variant name |

## Related Work

| System | Vend. | Meta | Pol. | Data | Ups. |
|---|---|---|---|---|---|
| IREE | ● | ● | ● | ● | ● |
| chipStar | ● | ● | ● | ● | ● |
| Proteus | ● | ● | ● | ● | ● |
| liboffload | ● | ● | ● | ● | ● |
| CPU FMV | ● | ● | ● | ● | ● |
| Ours | ● | ● | ● | ● | ● |

● Yes ● Partial ● No

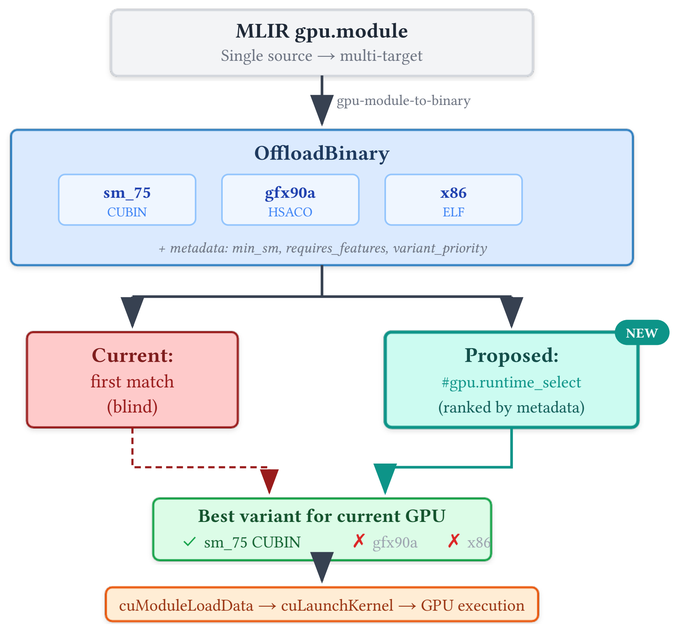
First to combine all five.

## The Insight

Since selection costs **3–6 ns**, the question is not *whether* to dispatch, but **what information drives it**. We formalize kernel variant selection as a **multi-armed bandit** problem.

Phase 1 measured the dispatch. Phase 2 makes it **intelligent**.

## System Architecture




End-to-end: MLIR embeds metadata in OffloadBinary; at runtime, the MAB profiler scores each variant and dispatches the best match.

## The MAB Formulation

**Arms:** $N$ pre-compiled kernel variants (typically $N < 10$)
**Reward:** Negative execution time: $r_i = -t_{\text{exec}(v_i)}$
**Context:** $(kernel_name, shape, device_id) \rightarrow$ cacheable key
**Objective:** Minimize cumulative regret $R_T = \sum_{t=1}^{T} \left(\mu^* - \mu_{a_t}\right)$

**Structural Properties:**

| $N < 10$ | $\sigma^2 < 5\%$ | Cacheable |
|---|---|---|
| arms (few variants) | low noise | contexts reuse |

## Why This Is a Degenerate Bandit

Standard MAB algorithms (UCB1, Thompson Sampling) are designed for $N \rightarrow \infty$ or non-stationary rewards. GPU kernel dispatch has **none** of these complexities:

**Exhaustive exploration** ($N \times w$ warmup samples) followed by **permanent exploitation**.
**Regret:** $O(N^2)$ constant — provably optimal for this problem class.
**UCB1 and Thompson Sampling are unnecessary.**

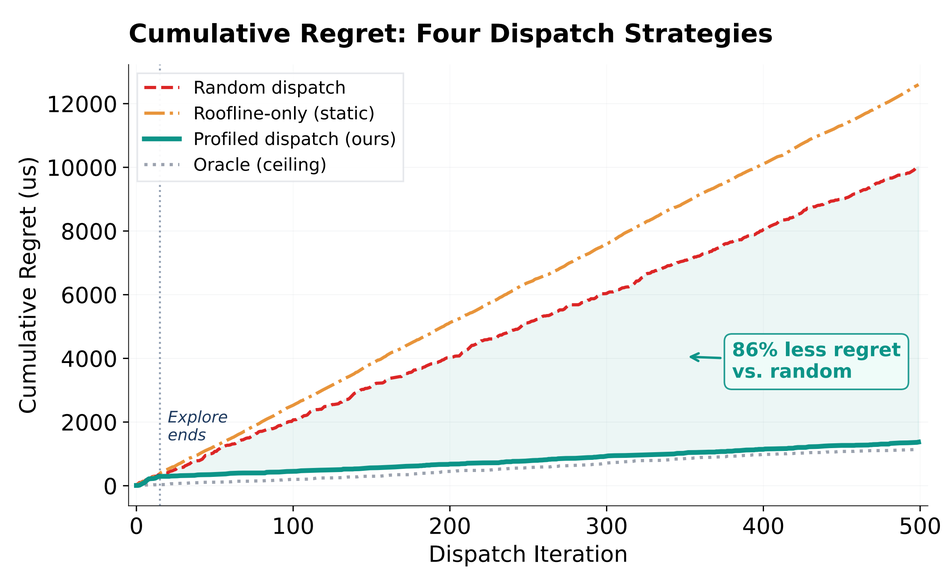
With $N = 3$ variants and $w = 3$ warmup rounds, convergence is guaranteed in **9 dispatches**. After that, every dispatch is optimal with zero marginal regret.

## The Algorithm

```
fn dispatch(ctx, variants[N], warmup=3):
  key = (ctx.kernel, ctx.shape, ctx.device)
  if key in cache:           // EXPLOIT
    return cache[key]
  if stats[key].count < N*warmup:  // EXPLORE
    arm = stats[key].count % N
    t = time(variants[arm])
    stats[key].update(arm, t)
    return variants[arm]
  cache[key] = argmin(stats[key].mean)  // LOCK
  return cache[key]
```

Three phases: **Cold** (no data) → **Explore** (round-robin warmup) → **Exploit** (permanent lock).

## Profiled vs. Baselines



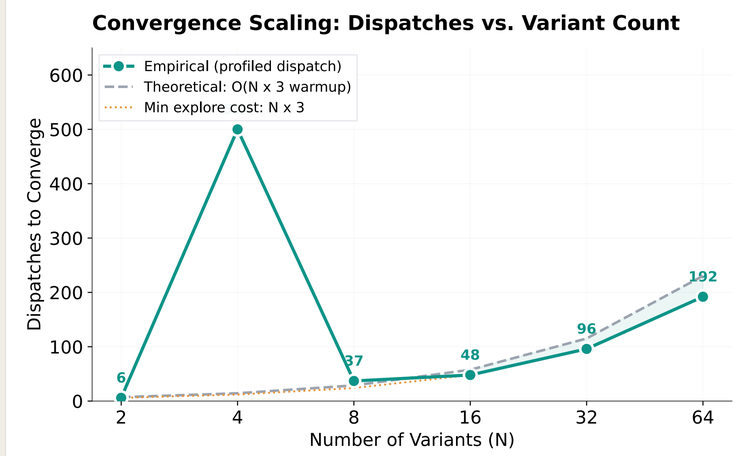

| 86% | O(N²) | 7.3x |
|---|---|---|
| Of oracle performance | Regret bound (constant) | Less regret than random |

*With 5 near-identical variants (12% spread, 8% noise), profiled dispatch converges and achieves near-oracle performance — 7.3x better than random, robust to noise.*

## Dispatch Latency Breakdown

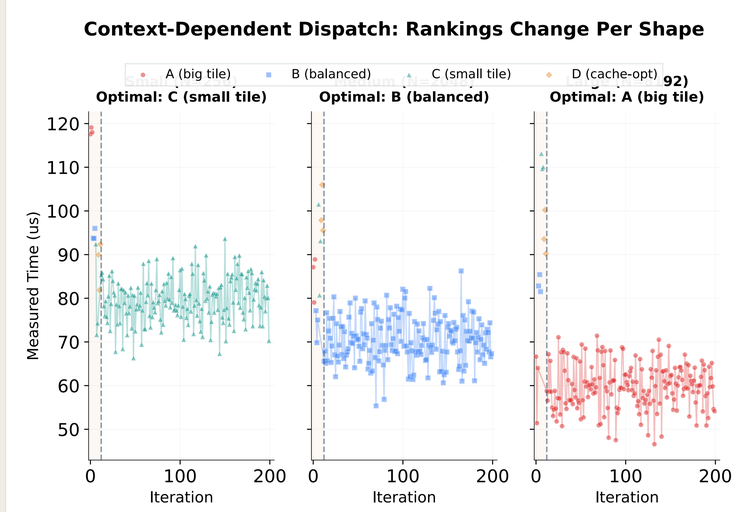
| Operation | Latency | Share |
|---|---|---|
| cuModuleLoadData (cold) | 36.0 µs | 89.6% |
| cuModuleLoadData (warm) | 9.6 µs | – |
| cuModuleGetFunction | 63 ns | 0.2% |
| cuLaunchKernel | 1.65 µs | 4.1% |
| cuStreamSynchronize | 2.45 µs | 6.1% |
| **Hot-path total** | **4.1 µs** | **launch+sync** |
| **Selection overhead** | **3–6 ns** | **< 0.02%** |

## Scaling with Variants



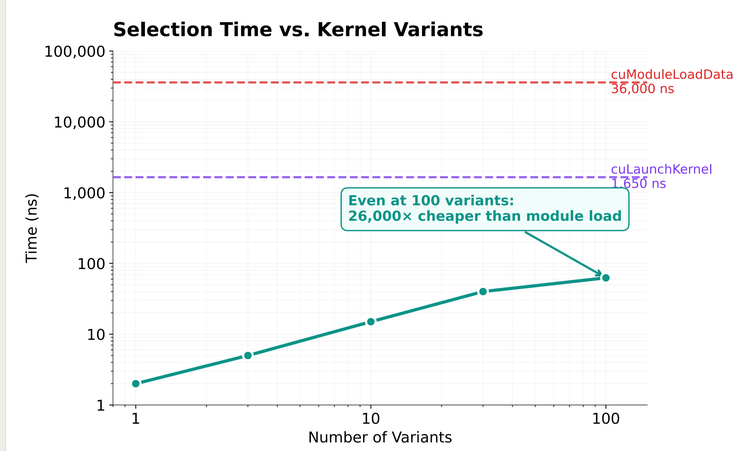

Profiled dispatch achieves 86% of oracle with 7.3x less regret than random.

## Context-Dependent Selection




Different shapes converge to different optimal variants — the context in contextual bandit matters.

## Selection Scales Linearly




Even at 64 variants, selection stays under 400 ns — three orders of magnitude below driver overhead.

## Prototype Implementation

| 5,157 | 664 | 238 |
|---|---|---|
| LOC — libkdl (dispatch lib) | LOC — PoC (MLIR integ.) | LOC — MAB profiler |

## Key Findings

**F1:** Module loading dominates cold dispatch at 90% — selection is essentially free.
**F2:** GPU dispatch is a *degenerate* bandit — exhaustive exploration provably optimal.
**F3:** Convergence in 9 dispatches; zero marginal regret after lock-in.
**F4:** Linear variant scaling works from 2 to 64+ variants without changes.

## Upstream Path

1. **Metadata RFC** — 5 keys, ~30 LOC patch
2. **Policy Slot** — Pluggable hook in liboffload
3. **MAB Profiler** — Drop-in profiled dispatch

*Each step independently useful.*

**Code:** github.com/Akasxh/libkdl · **Contact:** sakash@iitp.ac.in
**Refs:** Auer et al. (2002) UCB1 · SparseX, CGO 2026 · **EuroLLVM Dublin 2026**
**5,157 + 664 + 238 LOC** · Apache 2.0 w/ LLVM Exception